

India's First Hydrogen-Powered Mining Truck Ushers in a Clean-Energy Era

Adani Group's introduction of India's first hydrogen-powered truck for mining logistics marks a significant leap toward sustainable industrial transformation. Developed in partnership with auto maker and hydrogen fuel cell technology developer, the 55-tonne truck operates on green hydrogen, eliminating tailpipe emissions and significantly reducing the carbon footprint of heavy-duty mining logistics.

Traditionally, mining fleets run on diesel, contributing

substantially to greenhouse gas emissions. The hydrogen truck is a gamechanger—not only does it support India's decarbonization goals, but it also showcases Adani Group Chairman Gautam Adani's visionary leadership in integrating cutting-edge clean technology into core infrastructure sectors. With plans to scale hydrogen usage across other logistics and mining hubs, this innovation could drastically cut operating costs in the long term and inspire industry-wide adoption. Positioned at the intersection of

energy, mobility, and sustainability, the Adani hydrogen truck symbolises a future where industrial strength meets environmental responsibility—redefining how India mines, moves, and manages its natural resources.

The truck was officially commissioned on 10 May when Chhattisgarh's Chief Minister Vishnu Deo Sai flagged in the state capital Raipur for operations between the Gare Pelma-III coal block in Raigarh district and a nearby state-owned power plant.

aimed at reducing vibrations, fly rock, and emissions. It represents a paradigm shift in India's opencast mining operations.

Greening the Ground: Community-Led Environmental Restoration

Beyond tech innovation, the PEKB story is also about ecological and social regeneration. At mine's site, a high-tech nursery produces healthy saplings for large-scale afforestation. So far, over 15.69 lakh trees have been planted across 1,200 acres of reclaimed land, using native species like sal, creating self-sustaining green belts and restoring biodiversity.

Adani Natural Resources led a record-breaking plantation of 27,220 saplings on 5 June, which is celebrated as World Environment Day, across PEKB and Parsa Coal Blocks that are owned by Rajasthan Government's Power Generation Utility that has awarded mine development and operation contract to Adani Enterprises Limited. Students, villagers, contractors, and employees participated in activities such as drawing competitions themed around "Beat Plastic Pollution", turning awareness into grassroots action.

Similar green initiatives are underway at Gare Pelma-III (GP-III) in Raigarh district, which is owned by the Chhattisgarh government and operated by AEL, focusing on eco-restoration and low-impact mining practices.

Beyond the Mine: Social Commitments in Action

Adani Natural Resources' interventions go beyond ecology. In nearby communities, the group has been:

- Renovating government schools
- Building sanitation facilities
- Paving village roads and constructing bus shelters
- Providing educational support for 1,000+ children for free.

Together, these efforts reflect a holistic, future-facing model of mining that balances productivity with purpose, profit with people, and extraction with regeneration.

Impact on the Sector

The diversified conglomerate's innovation-first model is elevating standards across the industry. Officials of other mining companies visit to learn from Adani's mining model.

The Big Takeaway

In an industry long defined by environmental compromise and inefficiency, the Adani Group is proving that mining can be cutting-edge, clean, and community orientated.

Mines of the Mind: Where Automation Meets Earth

Adani Natural Resources is transforming mining with a three-pronged approach and how:

Smart Extraction Technologies

- Robotics-driven drilling and load and haul systems
- Real-time monitoring using state-of-the-art technologies

Eco-Conscious Operations

- Progressive reclamation: Land restored as green zones with native flora
- Shift to solar-powered site operations

Empowering Rural Youth Through Mining & Driving Training Initiatives

The role of a Mining Sirdar is critical to safe and efficient mining operations. Responsible for conducting inspections, monitoring ventilation systems, enforcing safety protocols, and mitigating potential hazards, the Sirdar serves as a frontline guardian of underground safety.

Recognising the importance of this role, Adani Natural Resources has launched a coaching programme to prepare rural youth for the Mining Sirdarship Certificate of Competency examination, administered by the Directorate General of Mines Safety (DGMS). The initiative has already shown encouraging results, inspiring many youths—especially those from underserved communities and with only Grade XII education—to pursue a career path once thought out of reach. This opportunity is opening doors to stable employment and helping uplift the living standards of their families.

In a parallel effort to enhance employability, Adani Skills and Education has partnered with the Institute of Driving and Traffic Research (IDTR) in Chhattisgarh to launch a free professional training programme. This initiative includes 240 hours of Heavy Motor Vehicle (HMV) training and 200 hours of Heavy Earth Moving Machinery (HEMM) training. The programme is open to youth from Chhattisgarh, Madhya Pradesh, and Odisha, equipping them with industry-relevant skills for better job prospects in logistics and mining sectors.



The Dozer Push Mining Method introduces automated, unmanned mining technology to enhance safety, reduce environmental impact, and increase operational efficiency.